

1.Q: What do you know about the fundamental concept of Bohr's atomic model?

Ans:

The Bohr Model

- The main fundamental concept of Bohr's model is that electrons in an atom can only **occupy certain discrete energy levels**
- Electrons **can move one energy level to another energy level** by absorbing or emitting energy in the form of photons, which are packets of electromagnetic radiation. When an electron absorbs a photon, it gains energy and moves to a higher energy level. When an electron emits a photon, it loses energy and moves to a lower energy level.
- The Bohr model **also introduced the concept of the principal quantum number**, which determines the energy level of an electron. The model also explained the **spectral lines**.
- Bohr proposed that an electron is found only in specific circular paths, or orbits, around the nucleus.
- Each possible electron's orbit in Bohr's model **has a fixed energy**. The fixed energies an electron can have are called energy levels.
- The energy levels get lower in energy as you move closer towards the nucleus.
- The energy levels also get higher in energy as you move farther from the nucleus.
- Niels Bohr studied the hydrogen atom because it was the most simplistic.

***Bohr's Model Restrictions**

Bohr's model accurately describes the movement of an electron in the hydrogen atom, but it cannot describe the movement of **multi-electron atoms**.

2.Q: Cite two important quantum mechanical concepts associated with the Bohr model of the atom.

Ans:

The Bohr model of the atom, proposed by Niels Bohr in 1913, is a seminal work in the history of quantum mechanics. Two important quantum mechanical concepts associated with the Bohr model of the atom are:

- ☐ **Quantization of energy:** The Bohr model explained that electrons in an atom can only occupy certain fixed energy levels or orbits, and can jump from one energy level to another by emitting or absorbing a specific amount of energy. This quantization of energy was a revolutionary concept at the time, and it laid the foundation for the development of quantum mechanics.

- ☐ **Wave-particle duality:** The Bohr model suggested that electrons exhibit both wave-like and particle-like behavior, which is known as wave-particle duality. This concept was later elaborated upon in quantum mechanics, which showed that all matter exhibits wave-particle duality, not just electrons.

3.Q: Give a comparison in Bohr and wave mechanical model of an atom in terms of electron distribution.

Ans:

The Bohr model of an atom was proposed by Niels Bohr in 1913, while the wave mechanical model was proposed by Erwin Schrodinger in 1926. Both models explain the electron distribution in an atom in different ways.

- In the Bohr model, electrons are assumed that electrons in an atom can only **occupy certain discrete energy levels**, meaning that electrons can only exist at certain discrete energy levels and not in between. Each orbit has a fixed radius and energy level, and the electrons do not radiate energy while they move around the nucleus. The electron distribution in the Bohr model is discrete, with each electron occupying a specific orbit around the nucleus.
- On the other hand, the wave mechanical model describes electrons as waves, rather than particles orbiting the nucleus. The electron's location can only be described in terms of its probability of being in a certain region around the nucleus, known as an electron cloud or an atomic orbital. The electron cloud represents the three-dimensional region around the nucleus where the electron is most likely to be found. The electron distribution in the wave mechanical model is continuous, with electrons occupying a range of positions and energies within the electron cloud.

In summary, the Bohr model describes the electron distribution in an atom in terms of discrete energy levels and orbits, while the wave mechanical model describes it in terms of a continuous probability distribution of electron density around the nucleus.

4.Q: Discuss different types of primary interatomic bonds in solids.

Ans:

There are three types of primary interatomic bonds that hold atoms together in a molecule or a solid: covalent, ionic, and metallic bonds.

- A covalent bond is a type of chemical bond formed between two non-metal atoms by sharing one or more pairs of electrons. In a covalent bond, the atoms share electrons in order to achieve a stable electron configuration. The atoms in the molecule are held together by the attraction between the positively charged

nuclei and the negatively charged shared electrons. Covalent bonds can be polar or nonpolar, depending on the electronegativity difference between the atoms. An example of a covalent bond is the bond between two hydrogen atoms in a hydrogen molecule (H_2). Each hydrogen atom has one electron in its outermost shell, and by sharing these electrons, they both achieve a stable electron configuration similar to that of helium (He). The shared pair of electrons forms a covalent bond that holds the two hydrogen atoms together in the molecule.

- An ionic bond is a chemical bond between a metal and a non-metal where electrons are transferred. The atom with the low ionization energy will lose one or more electrons to the atom with the high electron affinity, creating ions with opposite charges. This results in one positively charged ion (cation) and one negatively charged ion (anion) that are attracted to each other by electrostatic forces. Ionic bonds are typically found in compounds that contain a metal and a nonmetal.
- Metallic bonds occur when metal atoms share their valence electrons with neighboring atoms. In metallic bonding, the outermost electrons of metal atoms become delocalized, forming a sea of electrons that can move freely throughout the metal lattice. These delocalized electrons are responsible for the characteristic properties of metals such as high electrical and thermal conductivity, malleability, and ductility. The properties of metallic bonds include high electrical and thermal conductivity, malleability, ductility, and a tendency to form a crystal lattice structure. Metallic bonds are typically found in metals and alloys.

In summary, covalent bonds involve the sharing of electrons between atoms, ionic bonds involve the transfer of electrons from one atom to another, and metallic bonds involve the sharing of electrons among many atoms in a metal lattice.

5.Q: Write down the properties of ionic bond and covalent bond in solids.

Ans

Properties of ionic bonds in solids

i). High melting and boiling points: Ionic solids have high melting and boiling points due to the strong electrostatic attraction between the positively and negatively charged ions.

ii). Hardness and brittleness: Ionic solids are hard and brittle due to the rigid structure of the lattice, which can shatter upon impact.

iii). Poor electrical conductivity: Solid ionic compounds do not conduct electricity because the ions are held in fixed positions and cannot move freely to carry an electric charge.

iv). **Solubility in polar solvents:** Ionic compounds are generally soluble in polar solvents, such as water, because the polar molecules can attract and surround the charged ions and help them dissolve.

Properties of covalent bonds in solids

i). **Low melting and boiling points:** Covalent solids have low melting and boiling points because the intermolecular forces between the molecules are relatively weak.

ii). **Softness:** Covalent solids are often soft due to the weak intermolecular forces between the molecules.

iii). **Poor electrical conductivity:** Most covalent solids do not conduct electricity because they have no charged particles that can move freely to carry an electric charge.

iv). **Insolubility in polar solvents:** Covalent compounds are generally insoluble in polar solvents because the polar molecules cannot interact strongly enough with the nonpolar covalent molecules to dissolve them.

In summary, ionic bonds in solids are characterized by high melting points, hardness, brittleness, poor electrical conductivity, and solubility in polar solvents, while covalent bonds in solids are characterized by low melting points, softness, poor electrical conductivity, and insolubility in polar solvents.

6.Q: Symmetrically explain the metallic bonding.

Ans:

Metallic bonds occur when metal atoms share their valence electrons with neighboring atoms. In metallic bonding, the outermost electrons of metal atoms become delocalized, forming a sea of electrons that can move freely throughout the metal lattice. These delocalized electrons are responsible for the characteristic properties of metals such as high electrical and thermal conductivity, malleability, and ductility. The properties of metallic bonds include high electrical and thermal conductivity, malleability, ductility, and a tendency to form a crystal lattice structure. Metallic bonds are typically found in metals and alloys.

Because the electrons are free to move, metallic bonds have unique properties such as high electrical conductivity, thermal conductivity, and malleability. When a voltage is applied, the delocalized electrons can easily move through the metal lattice, allowing for the flow of electric current. When heat is applied, the delocalized electrons can transfer thermal energy quickly throughout the metal, resulting in high thermal conductivity.

Overall, metallic bonding is the result of the sharing of electrons among many metal atoms, creating a unique and strong bond that gives metals their characteristic properties.

Properties of metallic bond

High thermal conductivity: Metals have high thermal conductivity because of the free movement of electrons throughout the metal lattice. This allows for rapid transfer of heat energy through the metal.

High electrical conductivity: Metallic bonds also contribute to the high electrical conductivity of metals, as the free electrons can easily move through the lattice in response to an electric field.

Malleability and ductility: Metals are malleable and ductile, meaning they can be easily shaped and stretched without breaking. This is due to the ability of metal atoms to slide past each other while still maintaining the metallic bond.

High melting and boiling points: Metals generally have high melting and boiling points, which is attributed to the strong metallic bonds that hold the metal lattice together.

7.Q: Briefly gives differences between ionic, covalent and metallic bonding.

Ans:

The main differences between these three types of bonds are:

i). Electron sharing: In covalent bonds, electrons are shared between two or more atoms, while in ionic bonds, electrons are transferred from one atom to another. In metallic bonds, electrons are shared between many metal atoms.

ii). Bond strength: Covalent bonds are generally stronger than ionic bonds, which are stronger than metallic bonds. Covalent bonds involve sharing electrons, resulting in strong electrostatic forces holding the atoms together. Ionic bonds involve the attraction between oppositely charged ions, which are also strong but not as strong as covalent bonds. Metallic bonds involve a sea of delocalized electrons shared between many atoms, resulting in weaker forces compared to covalent and ionic bonds.

iii). Types of elements involved: Covalent bonds typically form between two nonmetal atoms, while ionic bonds typically form between a metal and a nonmetal atom. Metallic bonds form between metal atoms.

iv). Physical properties: The type of bond affects the physical properties of a substance. Covalent compounds tend to have low melting and boiling points, while ionic compounds have high melting and boiling points. Metals are good conductors of heat and electricity due to their metallic bonds.

V). Polarity: Covalent bonds can be polar or nonpolar, depending on the electronegativity difference between the atoms. Ionic bonds are always polar. Metallic bonds are typically nonpolar.

8.Q: Give the significance of each of the four Quantum numbers.

9.Q: Relative to electrons and electron states, what does each of the four quantum number specify?

Ans:

The four quantum numbers describe the state of an electron in an atom. Here's the significance of each of them:

i).Principal Quantum Number (n): This quantum number describes the energy level of an electron in an atom. It determines the size and the distance of the electron from the nucleus. The higher the value of n , the farther away the electron is from the nucleus and the higher its energy. The principal quantum number can have any positive integer value, such as 1, 2, 3, and so on.

ii).Azimuthal Quantum Number (l): This quantum number describes the shape of the electron cloud or the orbital in which an electron is located. It determines the angular momentum of the electron around the nucleus. The values of l depend on the value of n and range from 0 to $n-1$. For example, if $n=2$, l can have values of 0 or 1. If $n=3$, l can have values of 0, 1, or 2.

iii).Magnetic Quantum Number (m): This quantum number describes the orientation of the electron cloud or orbital in space. It determines the direction of the electron's angular momentum with respect to an external magnetic field. The values of m depend on the value of l and range from $-l$ to $+l$. For example, if $l=1$, m can have values of -1, 0, or 1.

iv).Spin Quantum Number (s): It describes the intrinsic angular momentum or spin of the electron around its own axis. The value of ' s ' can be either $+1/2$ or $-1/2$, corresponding to the two possible spin states of an electron. This quantum number distinguishes one electron from another within an orbital.

Together, these four quantum numbers give a complete description of the state of an electron in an atom, and they are essential for understanding atomic structure and chemical bonding.

Together, these quantum numbers specify the energy, shape, orientation, and spin state of an electron in an atom. By using these quantum numbers, we can determine the probability of finding an electron in a particular region around the nucleus, the allowed energy levels, and the chemical behavior of atoms in chemical reactions.

10.Q: Distinguish between alloy and intermetallic compound.

Ans:

Both alloys and intermetallic compounds involve the combination of two or more metals. However, there are important differences between them:

i). Composition: Alloys typically contain one or more metals as well as non-metallic elements (such as carbon in steel), whereas intermetallic compounds consist solely of two or more metals combined in a fixed stoichiometric ratio.

ii). Atomic structure: Alloys are typically composed of randomly arranged atoms of different metals, whereas intermetallic compounds have a well-defined, ordered atomic structure.

iii). Properties: Alloys generally have properties that are intermediate between those of the constituent metals, such as increased strength, hardness, or resistance to corrosion. Intermetallic compounds, on the other hand, often exhibit unique properties that are not found in the constituent metals or alloys, such as high melting points, high-temperature stability, and low ductility.

iv). Formation: Alloys can be formed by melting and mixing metals together, or by coating one metal onto another. Intermetallic compounds, in contrast, typically require specific conditions of temperature, pressure, and composition to form, and often require a complex sequence of reactions to synthesize.

In summary, while both alloys and intermetallic compounds involve the combination of metals, they differ in their composition, atomic structure, properties, and formation process.

11.Q: Upon freezing most substances experience a decrease in volume but opposite happens in case of water explain why?

Ans:

Most substances do indeed experience a decrease in volume upon freezing because the molecules become more ordered and arranged in a regular crystalline lattice structure. However, water is an exception to this rule. When water freezes, it actually expands and increases in volume. This is because of the unique way in which the hydrogen bonds between water molecules form and interact with each other.

In liquid water, the hydrogen bonds between the molecules are constantly breaking and reforming, allowing the molecules to move and flow relatively freely. However, when the temperature drops and water begins to freeze, the molecules become less mobile and the hydrogen bonds become more stable. As a result, the water molecules arrange themselves into a rigid, crystalline structure that has a higher volume than the liquid.

Moreover, the hydrogen bonds in ice are arranged in a way that creates empty spaces or voids in the crystal structure. These spaces cause the ice to be less dense than liquid water, leading to its expansion upon freezing. This is why ice floats on liquid

water, and why lakes and rivers freeze from the top down, allowing aquatic life to survive beneath the frozen surface.

In summary, the expansion of water upon freezing is due to the unique way in which hydrogen bonds form and interact with each other, creating a crystalline structure with empty spaces that increases the volume of the frozen water.

12. Q: Explain why HF has a higher boiling temperature than HCL even though HF has a lower molecular weight?

Ans:

Although hydrogen fluoride (HF) has a lower molecular weight than hydrogen chloride (HCl), it has a higher boiling temperature due to the presence of strong intermolecular hydrogen bonds between the HF molecules. These hydrogen bonds are formed between the highly electronegative fluorine atom in one molecule and the hydrogen atom in a neighboring molecule.

i). The strength of the hydrogen bonds depends on the difference in electronegativity between the hydrogen and the other atom, as well as the distance between the atoms. In the case of HF, the electronegativity difference between hydrogen and fluorine is large, and the molecule has a relatively small size. As a result, the hydrogen bonds between HF molecules are very strong, which requires a lot of energy to break and causes HF to have a higher boiling temperature.

ii). In contrast, HCl has weaker intermolecular forces due to the smaller electronegativity difference between hydrogen and chlorine, and the larger size of the chlorine atom compared to the fluorine atom. This results in weaker dipole-dipole interactions between HCl molecules, which require less energy to overcome, causing HCl to have a lower boiling temperature compared to HF.

In summary, even though HF has a lower molecular weight than HCl, its higher boiling temperature is due to the strong intermolecular hydrogen bonds between the HF molecules, which require more energy to overcome compared to the weaker intermolecular forces between HCl molecules.

Math

13 Q: Calculate the force of attraction between Ca^{2+} and an O^{2-} ion the centres of which are separated by a distance of 1.25 nm

We are given the following data:

- The distance between center of Ca^{2+} and O^{2-} is: $x_o = 1.25 \text{ nm} = 1.25 \times 10^{-9} \text{ m}$.

The value of a singly charged electron and proton is:
 $q = 1.6 \times 10^{-19} \text{ C}$.

The number of ion charges in Ca^{2+} and O^{2-} is:
 $N = 2$.

The value of charge of Ca^{2+} is: $q_o = Nq$.

The value of charge of O^{2-} is: $q_p = Nq$.

The expression for the electrostatic force of attraction between the Ca^{2+} and O^{2-} is:

$$F_o = \frac{kq_oq_p}{x_o^2}$$

$$\begin{aligned} &= \frac{9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2 \times (2)^2 \times (1.6 \times 10^{-19} \text{ C})^2}{(1.25 \times 10^{-9} \text{ m})^2} \\ &= 5.898 \times 10^{-10} \text{ N} \\ &\approx 5.90 \times 10^{-10} \text{ N} \end{aligned}$$

Thus the electrostatic force of attraction between the Ca^{2+} and O^{2-} is **$5.90 \times 10^{-10} \text{ N}$** .

The force of repulsion is equal to the electrostatic force between the Ca^{2+} and O^{2-} at the same distance.

Thus the electrostatic force of repulsion between the Ca^{2+} and O^{2-} is **$5.90 \times 10^{-10} \text{ N}$** .

14 Q:

QUESTION

How many grams are there in one amu of a material?

ANSWER

In one amu of a material there are 1.66×10^{-24} grams.